5.8.1 Transistor as a Switch

Transistor as a switch is a very important and useful application.

Transistor as a Switch – ON

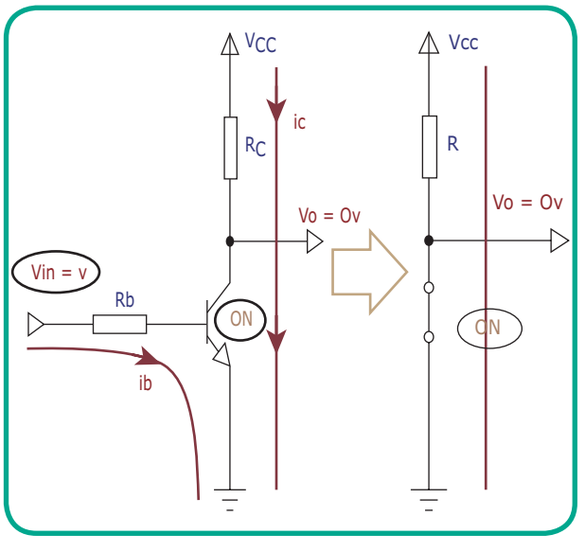


Figure 5.10 Transistor as a switch – ON Position

Figure 5.10 shows transistor as a switch (ON position). Transistor will become ON (saturation), when a sufficient voltage V is given to the input. During this condition, the collector-emitter voltage V_{CE} will be approximately equal to zero, i.e., the transistor acts as a short-circuit. For a silicon transistor, it is equal to 0.3 V.

Thus, the collector current $I_C = V_{CC}/R_E$ will flow through the circuit.

Transistors as a Switch – OFF

Figure 5.11 shows transistor as a switch OFF position. Transistors will be in OFF (cut off), when the input (V_{in}) is equal to zero. During this state, the transistor acts as an open- circuit and thus the entire voltage V_{CC} will be available at the collector.

5.8.2 Transistor as an Amplifier

We can use a transistor as an amplifier for increasing the strength of the weak

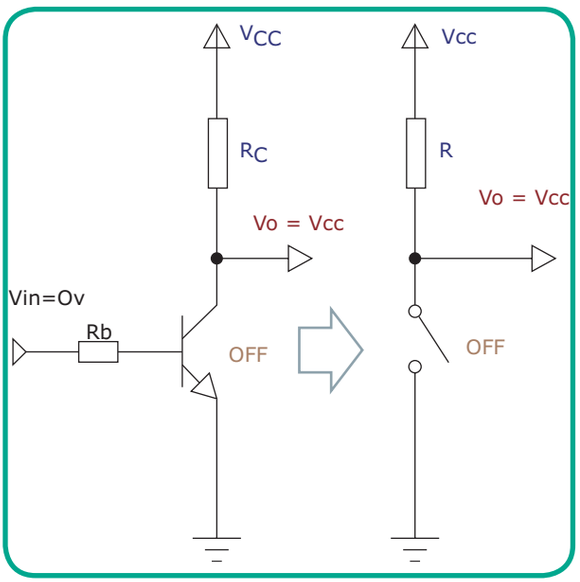


Figure 5.11 Transistor as a switch – OFF Position

signal. With the help of circuit diagram, we explain how a transistor acts as an amplifier.

Figure 5.12 shows the basic circuit of a transistor amplifier. The weak signal to be amplified is applied as the input signal between the emitter-base junction and the output is taken across the load R_L. For faithful amplification, the input circuit is forward biased and the output circuit is reverse biased. For this purpose, we apply dc voltage.

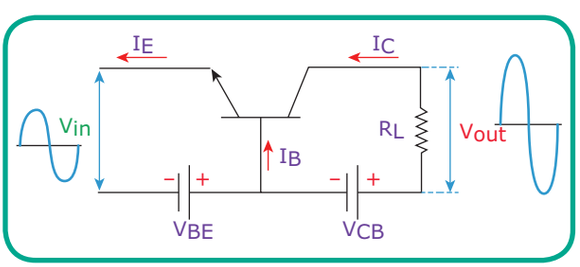


Figure 5.12 Basic Circuit of a transistor amplifier

As we know, the input circuit has low resistance; consequently, a small change in the signal voltage occurs at the input circuit lead to an appreciable change in the emitter current. Due to the transistor action, change in emitter current

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